Model metrics
Computation metrics -

College of Engineering
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Department of Artificial Intelligence and Machine Learning

Date: 13/1/2026	Improvement Test	Max. Marks:50
Semester: V	UG	Duration: $1\frac{1}{2}$Hrs
Course Title: Machine Learning Operations [MLOps]		Course Code:AI254TA

Sl. no.	QUESTIONS – PART B	M	BT	CO
1.	Explain the feedback loop in MLOps. Describe its stages and role in improving model performance over time.	6	2	4
2	A fintech startup launches an AI-powered loan approval system that uses customer financial history, spending behaviour, and demographic data to predict creditworthiness. After deployment, customers begin reporting concerns including: a. Rejections that seem biased against certain groups b. Lack of explanation on why a loan was denied c. Fear of data misuse or sharing without consent d. No visible channel for correction or appeal e. System relies only on historical bank records, leading to poor performance on new customer profiles With respect to this scenario: i. Identify and explain the key elements of Responsible AI that the company must address. ii. With reference to the above scenario, explain how governance frameworks ensure the model is deployed ethically and remains compliant with regulatory requirements.	10	3	5
3a	Mention the two sources of feedback data in deployed ML models.	4	2	3
3b.	Explain how Amazon Cloud services can be used to integrate machine learning models into a production system. Support your explanation with a neat and labelled architectural diagram.	6	3	3
4.	Explain the role of monitoring and observability in cloud-based MLOps. How do logging practices, particularly in Python, contribute to identifying and addressing model performance issues?	10	4	3
5	An e-commerce retailer is experiencing low customer engagement, high cart abandonment rates, and difficulty helping shoppers discover relevant products. The company decides to implement a machine learning–based recommendation system to personalize product suggestions across its website and mobile app. Using this scenario, answer the following: i. Identify the major business challenges that the retailer is facing and explain how machine learning models could address them. (4 Marks) ii. Draw the MLOPs pipeline architecture and explain how MLOps pipeline can support model development by enabling experiment	14	3,4	4

(15)

		tracking, version control and reproducibility. (2 Marks – diagram + 4 Marks explanation)			
	iii.	What are the different MLOPs tools you would choose for the MLOPs development and justify the selection (4 Marks)			